

一个具有Vasile取等的根式不等式

设 $x, y, z > 0$. 则

$$\sum_{\text{cyc}} \sqrt{\frac{1}{7} \left(\frac{43y}{x} + 12 \right)} \geq \begin{cases} 10 - \frac{43}{21} \frac{\sum (7x^3 - 97x^2y) + 512xyz}{43 \sum x^3 - 68xyz} \\ 10 - \frac{215}{3801} \frac{\sum (7x^3 - 97x^2y) + 512xyz}{\sum x^3} \end{cases}$$

证明

利用 Holder 不等式

$$\left[\sum_{\text{cyc}} \sqrt{\frac{1}{7} \left(\frac{43y}{x} + 12 \right)} \right]^2 = \left(\sum_{\text{cyc}} \sqrt{\frac{43y + 12x}{7x}} \right)^2 \geq \frac{[\sum (43y + 12x)(7x^2 + 14xy + 7y^2 + 40xz + 85yz + 28z^2)]^3}{\sum 7x(43y + 12x)^2(7x^2 + 14xy + 7y^2 + 40xz + 85yz + 28z^2)^3}$$

该放缩可以解决两个不等式.

$$\begin{aligned} & 1 \quad ((12*y + 43*z)*(28*x^2 + 40*x*y + 7*y^2 + 85*x*z + 14*y*z + 7*z^2) + \\ & 2 \quad (43*x + 12*z)*(7*x^2 + 85*x*y + 28*y^2 + 14*x*z + 40*y*z + 7*z^2) + \\ & 3 \quad (12*x + 43*y)*(7*x^2 + 14*x*y + 7*y^2 + 40*x*z + 85*y*z + 28*z^2))^3 / \\ & 4 \quad (7*y*(12*y + 43*z)^2*(28*x^2 + 40*x*y + 7*y^2 + 85*x*z + 14*y*z + 7*z^2)^3 \\ & + \\ & 5 \quad 7*z*(43*x + 12*z)^2*(7*x^2 + 85*x*y + 28*y^2 + 14*x*z + 40*y*z + 7*z^2)^3 \\ & + \\ & 6 \quad 7*x*(12*x + 43*y)^2*(7*x^2 + 14*x*y + 7*y^2 + 40*x*z + 85*y*z + 28*z^2)^3) \\ & - \\ & 7 \quad (10 - (43*(7*x^3 - 97*x^2*y + 7*y^3 + 512*x*y*z - 97*y^2*z - 97*x*z^2 + \\ & 7*z^3)))/ \\ & 8 \quad (21*(-68*x*y*z + 43*(x^3 + y^3 + z^3))))^2 \end{aligned}$$

$$\begin{aligned} & 1 \quad ((12*y + 43*z)*(28*x^2 + 40*x*y + 7*y^2 + 85*x*z + 14*y*z + 7*z^2) + \\ & 2 \quad (43*x + 12*z)*(7*x^2 + 85*x*y + 28*y^2 + 14*x*z + 40*y*z + 7*z^2) + \\ & 3 \quad (12*x + 43*y)*(7*x^2 + 14*x*y + 7*y^2 + 40*x*z + 85*y*z + 28*z^2))^3 / \\ & 4 \quad (7*y*(12*y + 43*z)^2*(28*x^2 + 40*x*y + 7*y^2 + 85*x*z + 14*y*z + 7*z^2)^3 \\ & + \\ & 5 \quad 7*z*(43*x + 12*z)^2*(7*x^2 + 85*x*y + 28*y^2 + 14*x*z + 40*y*z + 7*z^2)^3 \\ & + \\ & 6 \quad 7*x*(12*x + 43*y)^2*(7*x^2 + 14*x*y + 7*y^2 + 40*x*z + 85*y*z + 28*z^2)^3) \\ & - \\ & 7 \quad (10 - (215*(7*x^3 - 97*x^2*y + 7*y^3 + 512*x*y*z - 97*y^2*z - 97*x*z^2 + \\ & 7*z^3)))/ \\ & 8 \quad (3801*(x^3 + y^3 + z^3)))^2 \end{aligned}$$